Five-trial intervals and earlier evidence

Main comparison at the 150 ms scoring cutoff

Values are hybrid minus First-Fit in percentage points, with paired 95% intervals. The bootstrap resamples cases after aggregating the five trials within each case. W/T/L is based on case means.

Family	Mean gap and interval	Best gap and interval	Hybrid W/T/L
All	+0.48 [-0.32, +1.26]	+0.46 [-0.14, +1.14]	11 / 12 / 7
Geometry	-0.61 [-2.41, +1.19]	+0.06 [-1.14, +1.61]	3 / 2 / 5
Payload	-0.00 [-0.01, +0.00]	+0.00 [+0.00, +0.00]	0 / 9 / 1
Mixed	+2.06 [+0.99, +3.12]	+1.31 [+0.19, +2.73]	8 / 1 / 1

Earlier three-trial budget sweep

This historical run uses the same cases with three seeds. The five-second subset uses nine cases. The four-attempt cap is a reproducible setting and the attempts do different work per method.

Limit	First-Fit	Hybrid	Mean attempts FF / hybrid
Four attempts	67.35%	70.01%	3.5 / 3.5
150 ms	69.17%	69.40%	255.2 / 2.6
500 ms	69.77%	70.57%	856.4 / 8.8
1.5 s	70.35%	70.99%	2,564.2 / 26.1
5 s subset	72.06%	72.85%	7,726.1 / 73.7

The following pages retain the mixed-pool, explicit-QUBO and full-support checks as supporting evidence.

Mixed pools and candidate coverage

Time-limited HiGHS on mixed cases

Limit	HiGHS mean	Hybrid mean	On-time HiGHS	Proven on time
150 ms	26.21%	82.40%	30/30	4
500 ms	84.17%	85.23%	30/30	17
1.5 s	84.32%	86.10%	30/30	30
5 s subset	83.94%	86.41%	9/9	9

HiGHS builds one pool per trial. Its mean includes the empty fallback when no candidate result finishes on time. A certified optimum applies to that pool only. The hybrid can try multiple pools within a deadline.

Original mixed-pool diagnosis

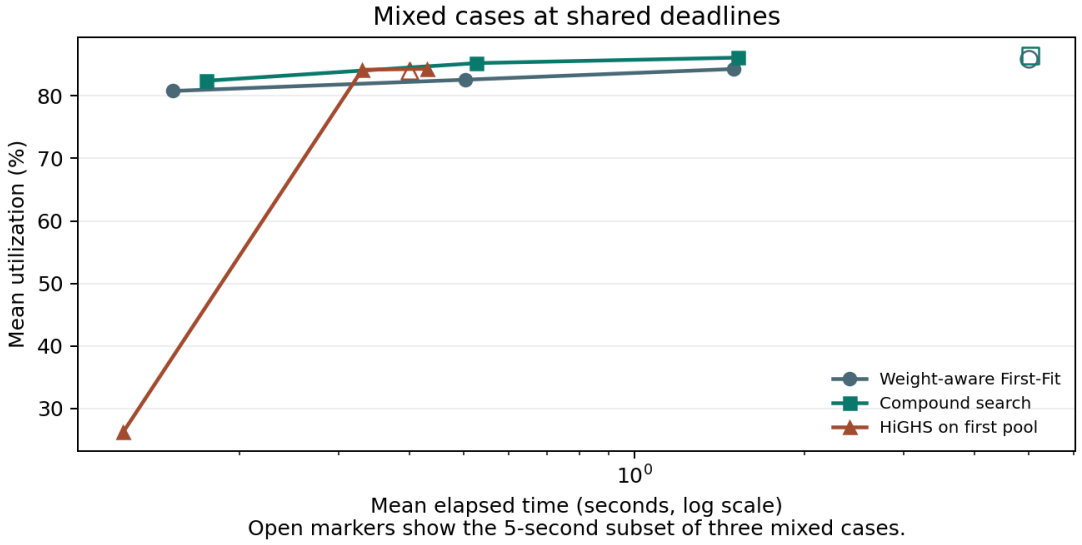
Across 30 recorded mixed pools, HiGHS proved 85.62% mean utilization versus 83.41% for the hybrid. The hybrid matched 10 of the 30 optima. The 2.21-point gap was available to better selection on those fixed pools. The 98.59% weight-and-volume bound ignores geometry.

Original geometry-pool diagnosis

Across 30 earlier geometry pools, candidate coverage accounted for 4.55 points of the gap to a full tiling. Search accounted for 0.21 points. In a three-case sensitivity check, varying corner priority raised one 20-box pool from 91.25% to 100%. This follow-up enables the change across all families.

Payload control

Knapsack followed by First-Fit matched the 28.02% payload upper bound on all ten payload cases. The same control averaged 70.26% on mixed cases. The payload family shows the benefit of choosing boxes by weight. The mixed family also requires the selected boxes to fit together.



Explicit QUBO and full support

Penalty sensitivity

Family	Original penalty	Best tested	Compound	Valid raw reads
Payload	25.83%	32.15%	32.15%	53/640
Mixed	53.54%	85.38%	86.01%	321/768

Values are mean best utilization after repair across three pools per family. The grid varied the squared slack penalty and used exact weight units or 1 kg units. Best tested uses the best setting for each pool and is an exploratory upper estimate. It does not measure how a frozen tuning rule generalizes.

The largest coefficient can dominate a box reward. This comparison changes that ratio and reports both raw validity and repaired volume. A weaker penalty can improve the search signal while making capacity violations cheaper.

Support rules

Original support-aware run	First-Fit	Hybrid	Gap (points)
Geometry	98.14%	91.33%	-6.81
Payload	27.71%	27.99%	+0.28
Mixed	78.03%	79.58%	+1.55

Checking support during search avoids losing a box when its base loses contact. The post-repair-only mode removes unsupported boxes and does not refill freed payload. This is a contact-area rule. Load-bearing strength and physical stability are separate engineering checks.